DECISION DEBT OS

Turning Business Decisions Into Measurable Learning

B.Tech Artificial Intelligence & Machine Learning · CA2 Capstone Project · AY 2026–27

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OBJECTIVE

• **Specific problem statement:** Businesses make repeated, consequential decisions — retention discounts, pricing changes, feature launches — without recording a measurable assumption about the expected outcome, and without later checking whether that assumption held true.

This creates **Decision Debt** — an accumulation of unresolved assumptions, overdue reviews, invalidated assumptions still driving active work, and repeated forecasting errors. Ordinary dashboards and project-management tools report what happened; none of them capture what a team believed would happen before it acted, so the belief can never be systematically checked.

Goal: build a lightweight decision-to-outcome tracking workflow — Decision Debt OS — that converts a business decision into a structured, measurable commitment and automatically evaluates it against reality.

LITERATURE REVIEW

- Discussing drawback of atleast 3 research papers (existing technology)

AUTHOR(S) & YEAR	TITLE	PROS	CONS	DIFFERENTIATION FROM DECISION DEBT OS
Scheibel & Rinderle-Ma (CAiSE 2023)	An End-to-End Approach for Online Decision Mining and Decision Drift Analysis	Monitors decision rules at runtime (not just after the fact); detects “decision drift”; validated on 4 synthetic + 1 real-life log	Tracks how an automated decision RULE changes — not whether a human's stated assumption about an outcome came true; needs existing event logs	We track a human assumption vs. a measured outcome; they track automated rule drift inside a process log — adjacent problem, different object
Goossens, De Smedt & Vanthienen (2024, KU Leuven)	Extracting Process-Aware Decision Models from Object-Centric Process Data	First object-centric decision-mining algorithm (IODDA); discovers decision structure + rules directly from logs; open benchmark released	Purely retrospective — reconstructs a model from data that already exists; can't recover an assumption that was never logged; no deadline/validation concept	We capture the assumption BEFORE the outcome is known (forward-looking); they mine structure AFTER the fact (backward-looking) — complementary, not a substitute
Natarajan, Mathur, Sidheekh, Stammer & Kersting (AAAI-25)	Human-in-the-loop or AI-in-the-loop? Automate or Collaborate?	Directly tackles when to automate a decision vs. keep a human in control; argues evaluation over-weights the model, under-weights the human expert	A conceptual/positional framework, not a deployable system; no mechanism to record or later validate the human's stated assumption	We operationalise exactly this division of labour: the model scores risk, a human approves + states the assumption, the system checks the outcome later
Gruetzemacher, Lee & Paradice (2024)	Calibration Training for Improving Probabilistic Judgments Using an Interactive App	Shows forecast calibration can be measurably improved with short, repeated feedback — supports asking for a stated confidence number	Small exploratory study (153 students, a sports-forecasting tournament); trains general forecasting skill, not tied to any business decision or metric	We don't train forecasting skill in the abstract — we capture a one-off, decision-specific forecast and check it against a real business metric

All four papers published 2023–2025 (peer-reviewed / top-tier preprint). Pros, Cons and Differentiation assessed against Decision Debt OS's specific mechanism — an explicit, falsifiable assumption bound to a deadline-evaluated outcome. Full citations on the References slide.

NOVELTY

• **What new techniques used:** Combines a churn-risk classifier (Random Forest, trained on real customer-behaviour data) with a rule-based Decision Evaluation Engine that checks a stated forecast against the actual outcome at a deadline, assigning one of six honest validation states — Validated, Partially Validated, Invalidated, Inconclusive, Stale, or Not Measurable.

• **How your topic is different from existing one:** BI dashboards report outcomes only; Jira/Notion capture decisions without a bound forecast; corporate prediction markets (P0, Hypermind) forecast a question but aren't tied to a company's own decision log; enterprise decision-intelligence patents are heavyweight, org-wide systems. Decision Debt OS is the first to combine an explicit, falsifiable assumption with automatic outcome evaluation in a lightweight workflow sized for a single product or retention team.

DATASET USED

• **Link:** Synthetic dataset generated for this project — no external public-repository link; file supplied directly as Customer_Transactions.csv.

• **Downloaded data set:** 10,000 fictional customer-decision records · 14 fields — age, gender, country, annual income, spending score, number of purchases, average purchase value, membership years, website visits/month, cart-abandon rate, churned (target), feedback text, last purchase date.

Note: *clearly labelled synthetic / fictional data, used for prototype demonstration only — not drawn from any real company's customers.*

ML ALGORITHM

• Implementation of problem formulation using ML

• **Problem formulation:** binary classification — predict whether a customer churns within 30 days (churned = 1) or is retained (churned = 0).

• **Features:** spending score, cart-abandon rate, website visits/month, number of purchases, average purchase value, membership years, annual income, age, gender, country.

• **Preprocessing:** StandardScaler on numeric features, One-Hot Encoding on categorical features (gender, country).

• **Train/test split:** 80% / 20%, stratified by class to preserve the churn ratio.

• **Algorithms compared:** Logistic Regression (baseline) vs. Random Forest Classifier, 300 trees, class_weight="balanced" to correct for churn being only 10.9% of the data.

IF IMPLEMENTED (RESULT)

• Which tools/language used to implement

Python 3 · Pandas · NumPy · Scikit-learn · Matplotlib

Result (Random Forest, held-out test set, n = 2,000) — real, computed values:

0.26	0.78	0.39	0.84
Precision	Recall	F1-score	ROC-AUC

Interpretation: optimised for recall over precision — catching 78% of true churners matters more than being right every time, since churn is only 10.9% of the data. The model flags risk; it does not prove any retention action caused a customer to stay — that requires a controlled comparison, which the Decision Evaluation Engine checks for.

CONCLUSION

• Decision Debt — the accumulation of decisions whose assumptions are never checked against reality — is a real, recurring gap that ordinary dashboards and project trackers do not close.

• Decision Debt OS prototypes a lightweight workflow (Capture → Structure → Forecast → Measure → Evaluate → Learn) that binds a stated assumption to a measurable metric and a deadline.

• On a real, if synthetic, 10,000-row dataset, risk-based targeting is justified: a high-risk cohort churns at 37.2% versus 1.5% for a low-risk cohort, and a Random Forest classifier reaches 0.84 ROC-AUC predicting that risk.

• The model predicts risk; it does not prove causation — human review and a controlled comparison remain required before crediting any intervention.

• Decision Debt OS does not replace human judgment. It makes the assumptions behind human decisions visible, measurable, and reviewable.

APPLICATIONS

• Customer-retention decisions — discounts, win-back offers, onboarding changes for at-risk customer segments

• Pricing and packaging decisions — testing a price change against a stated, measurable revenue/churn forecast

• Feature-launch decisions — tracking whether a shipped feature met its adoption or retention target

• Growth and marketing experiments — binding a campaign's expected lift to an explicit, checkable number

• Organisation-wide decision registries — a shared log of assumptions, forecasts and outcomes for audit and institutional learning

REFERENCES

Literature review (last 3 years)

1. Scheibel, B. & Rinderle-Ma, S. (2023). An End-to-End Approach for Online Decision Mining and Decision Drift Analysis in Process-Aware Information Systems. CAiSE 2023 Forum (LNBIP) / arXiv:2303.03961.

2. Goossens, A., De Smedt, J. & Vanthienen, J. (2024). Extracting Process-Aware Decision Models from Object-Centric Process Data. arXiv:2401.14847 (KU Leuven).

3. Natarajan, S., Mathur, S., Sidheekh, S., Stammer, W. & Kersting, K. (2025). Human-in-the-loop or AI-in-the-loop? Automate or Collaborate? Proc. AAAI-25, 39(27), 28594–28600.

4. Gruetzemacher, R., Lee, K.B. & Paradice, D. (2024). Calibration Training for Improving Probabilistic Judgments Using an Interactive App. Futures & Foresight Science, 6(2), e177.

Background & prior art

1. Organisational Memory and Decision Rationale — ScienceDirect

2. Operational Decision Discovery from Data — Springer

3. Improving Decisions with Market Information: Corporate Prediction Markets — Cambridge / Experimental Economics

4. Incentive-Compatible Forecasting — INFORMS / Management Science

5. P0 — Internal Prediction Markets (priorityzero.ai)

6. Hypermind — Crowd Forecasting (hypermind.com)

7. Vigil — Decision Tracking Use Cases (trustvigil.com)

8. Patent EP1402435A4 — Google Patents

9. Patent US20250217680A1 — Google Patents